

Quadratic Equations

1. Definition

A quadratic equation is a polynomial equation of the second degree.

Standard form: $ax^2 + bx + c = 0$ ($a \neq 0$)

2. Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

This is called the quadratic formula.

3. Discriminant

$$D = b^2 - 4ac$$

- $D > 0$: Two distinct real roots
- $D = 0$: One repeated real root
- $D < 0$: No real roots (complex)

4. Factoring Method

Example: $x^2 - 5x + 6 = 0$

Factor: $(x - 2)(x - 3) = 0$

Solution: $x = 2$ or $x = 3$

5. Completing the Square

Example: $x^2 + 6x + 5 = 0$

Step 1: $(x^2 + 6x + 9) - 9 + 5 = 0$

Step 2: $(x + 3)^2 = 4$

Step 3: $x + 3 = \pm 2$

Step 4: $x = -1$ or $x = -5$

6. Common Misconceptions

- Forgetting that a cannot be zero
- Sign errors when applying the quadratic formula
- Confusing $x^2 - 9 = 0$ with $x^2 + 9 = 0$
- Not checking discriminant before solving

- Dropping the +/- when taking square roots

7. Practice Problems

Q1: Solve $x^2 - 7x + 12 = 0$

Q2: Find discriminant of $2x^2 + 3x + 5 = 0$

Q3: Solve $3x^2 - 12 = 0$

Q4: Complete the square for $x^2 - 4x + 1 = 0$